

Day 22 Problem Set

Geometry Summer Credit | Scale Effects, Density, and Design Constraints

Name:	Date:	Period:
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Directions

State whether the problem uses length scale, area scale, volume scale, density, or a design constraint.

Part A: Scale Effects

1.	Two similar polygons have scale factor 5 from smaller to larger. The smaller perimeter is 12. Find the larger perimeter.
2.	Two similar figures have scale factor 3. The smaller area is 14 square units. Find the larger area.
3.	Two similar solids have scale factor 4. The smaller volume is 6 cubic units. Find the larger volume.
4.	A model car is built at scale factor $\frac{1}{10}$ of the actual car. If the actual car is 180 inches long, how long is the model?
5.	A square has side length 7. A similar square has side length 21. Find the scale factor and the area ratio.
6.	A cube's side length is tripled. What happens to its surface area? What happens to its volume?
7.	Error analysis: A student says if the scale factor is 2, then the volume doubles. Explain the error.

Part B: Density and Design Constraints

8. A solid has mass 500 g and volume 125 cm^3 . Find its density.

9. A metal cube has side length 4 cm and density 7 g/cm^3 . Find its mass.

10. A storage bin is 12 in by 10 in by 8 in. Does it meet a minimum capacity of 900 cubic inches?

11. A designer wants to reduce material by lowering surface area. Which measurement should be minimized: surface area or volume? Explain.

12. A cylinder has radius 3 cm and height 10 cm. Find its volume in terms of π .

13. If the cylinder in item 12 is filled with a liquid of density 2 g/cm^3 , write the mass in terms of π .

14. A package can weigh no more than 2000 g. A solid object has volume 300 cm^3 and density 8 g/cm^3 . Does it meet the weight constraint?

Day 22 Problem Set

Continued | Mixed Review

15.	Spiral review: Find the surface area of a rectangular prism with length 9, width 5, and height 4.
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16.	Spiral review: A cone has radius 6 and height 9. Find its volume in terms of π .
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17.	Spiral review: A circle has radius 10 and central angle 54 degrees. Find the sector area.
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18.	Application: A company doubles all dimensions of a cylindrical can. How do the label area and can volume change?
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19.	Challenge: Two similar water tanks have scale factor 3. The smaller tank holds 40 gallons. How much does the larger tank hold? Explain why the answer is not 120 gallons.
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